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COMPENDIUM

Real-World Impact of AI in Health

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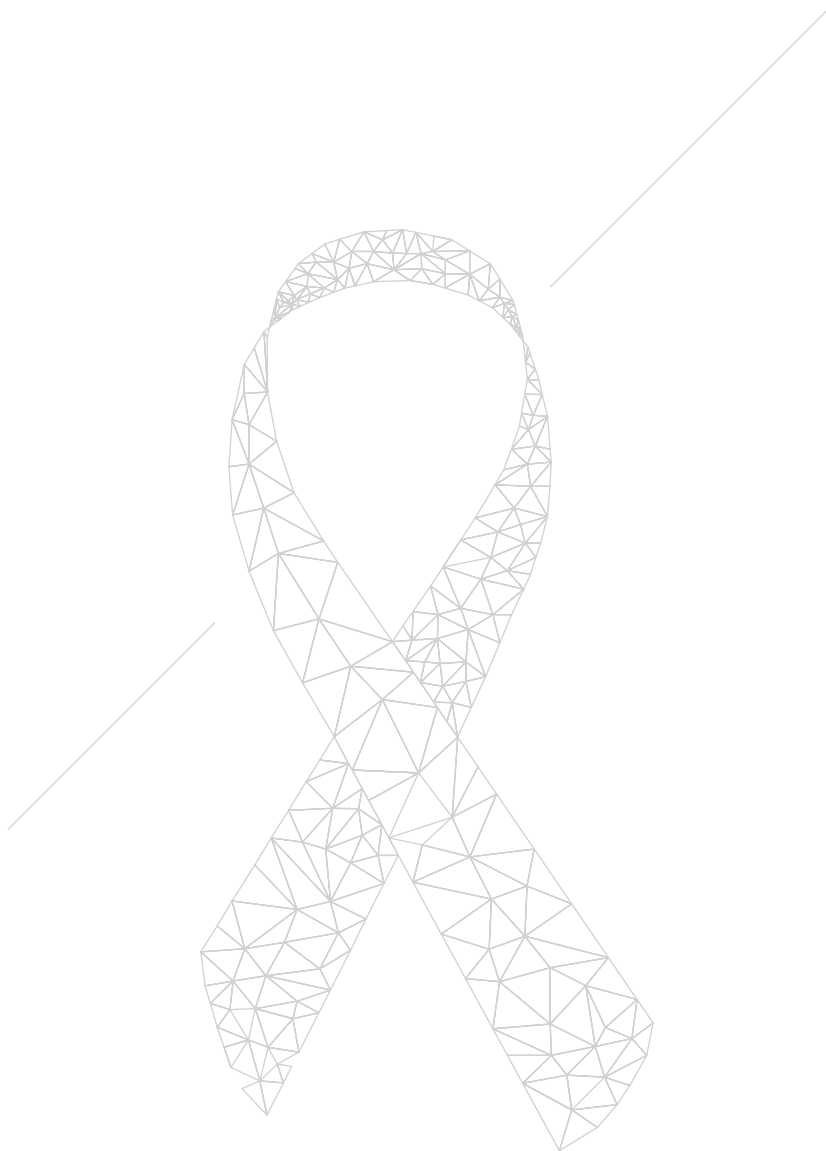
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Blueprint for getting AI ready data from Radiation Oncology: Insights from a large Indian tertiary care setting

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Abstract

Artificial intelligence is increasingly explored in medical imaging and oncology applications. However, in routine hospital environments particularly high-volume, technology-intensive departments such as radiation oncology, the primary constraint is often not algorithm availability but the presence of reliable, well-governed, and reusable clinical data. Without appropriate digital infrastructure, data stewardship processes, and workforce capacity, AI tools are difficult to develop, validate, and deploy responsibly.

This use case describes the experience of an Indian tertiary cancer centre in developing an AI-ready clinical and imaging data pipeline within a radiation oncology department. The work originated from a radiomics-based research question on recurrence prediction and evolved into a broader digital and governance transformation effort. The focus is on infrastructure, workflow redesign, and data stewardship rather than model performance.

Observed results relate primarily to infrastructure readiness, data completeness, governance maturity, and project enablement. The approach illustrates how AI readiness in clinical settings depends on interoperable systems, structured data capture, quality control, de-identification, secure storage, and human oversight. The framework is context-specific and is presented as an implementation blueprint that may inform similar efforts, with adaptation to local regulatory, governance, and infrastructure conditions.

Background

Radiation oncology departments rely on multiple digital and device-based systems for simulation, treatment planning, delivery, and follow-up. In settings with high patient volumes and constrained resources,

including many centres in India, operational pressures coexist with increasing interest in AI applications such as automated segmentation, planning support, and outcome prediction.

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A prospective observational study was initiated in 2020 to explore a radiomics-based model for predicting locoregional recurrence in head and neck cancer. The key question was asked, "Is this AI question relevant to our clinicians and patients, and if so, why should we pursue it?". An early feasibility assessment identified data fragmentation, missingness, and follow-up variability as limiting factors. This led to a reframing of priorities: rather than focusing first on model optimisation, the department prioritised building an AI-ready digital backbone and structured data pipeline embedded within routine care.

An interdisciplinary team including radiation oncologists, physicists, radiologists, data scientists, and IT specialists defined a patient-centred objective and recognised that sustainable AI efforts would depend on a redesigned data systems and governance processes rather than isolated model development.

AI ready use case

Solution

The initial modelling objective required access to planning CT images, clinical variables, treatment history, and longitudinal follow-up outcomes. Supervised models were first constructed using radiomic features describing image intensity, texture, and shape, linked with recurrence outcomes. Substantial data sparsity and missingness across key variables highlighted the need for an AI-ready data infrastructure capable of generating comprehensive, well-curated, and de-identified datasets aligned with data-centric best

practices for medical imaging and clinical data.

An end-to-end pipeline was therefore established with linked stages (1) patient identification and consent, (2) clinical data capture in the electronic medical record (EMR), (3) imaging acquisition and routing, (4) contouring and treatment planning, (5) automated and manual quality control (QC), (6) de identification and pseudonymisation, (7) secure storage and indexing, and (8) controlled access for analysis and model development (9) mitigation of longitudinal data loss during follow-up. Each stage was implemented through multidisciplinary coordination. The architecture leveraged existing enterprise clinical systems including the hospital information system (HIS), oncology information system (ARIA, Varian Medical Systems), treatment planning system (Eclipse, Varian Medical Systems), and Picture Archival and Communications Systems (PACS) platforms such as the open source Orthanc and XNAT to ensure AI readiness was embedded within routine clinical workflows (ARIA Oncology Information System | Varian, n.d.; Gutman et al., 2014; Jodogne, 2018; Marcus et al., 2007). Teleconsultation services aligned with national guidance (e-sanjeevani guidelines) were incorporated to support follow-up data capture (eSanjeevani, n.d.). The pipeline supports AI model development and evaluation but does not itself perform clinical decision-making.

Deployment and implementation

Digital transformation of radiation oncology records

Prior to 2019, the department used a hybrid information model combining institutional

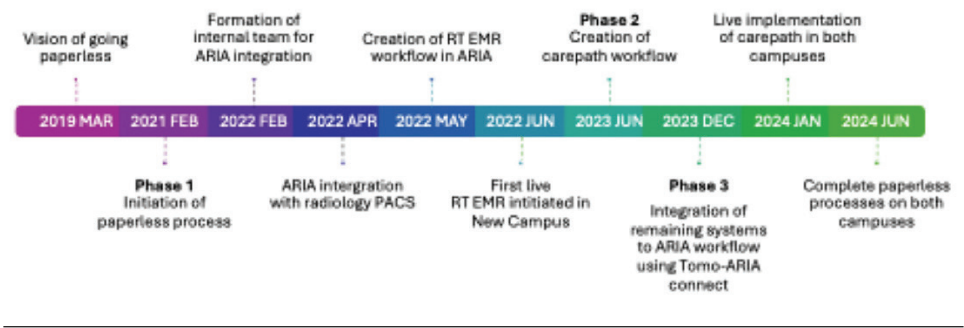
digital systems with radiation-oncology-specific paper charts. While functional in a single-campus setting, this model limited the data reuse and scalability and produced non-machine-readable records.

Beginning in 2021, the department transitioned to an integrated radiation oncology EMR linked to the hospital information system. A multidisciplinary mapping exercise translated all major paper-based touchpoints into structured electronic equivalents. The ARIA Oncology Information System was selected based on existing

licences, domain functionality, and remote access capability

Implementation followed a phased approach aligned with clinical operations (Figure 1). Phase 1 converted new-campus workflows to structured electronic capture. Phase 2 extended structured capture to the original campus with a temporary dual-system period until completeness and reliability were verified. Phase 3 connected treatment platforms not natively integrated with the treatment planning system through available connectivity modules.

Figure 01: Timeline of digital transformation of radiation oncology records



This transformation supported structured and interoperable data capture. It functions as enabling infrastructure rather than an AI model or decision system.

AI-ready data pipeline in routine care

Pipeline operationalisation was aligned with routine workflows so that data curation

emerged as a by-product of standard clinical processes (Figure 2).

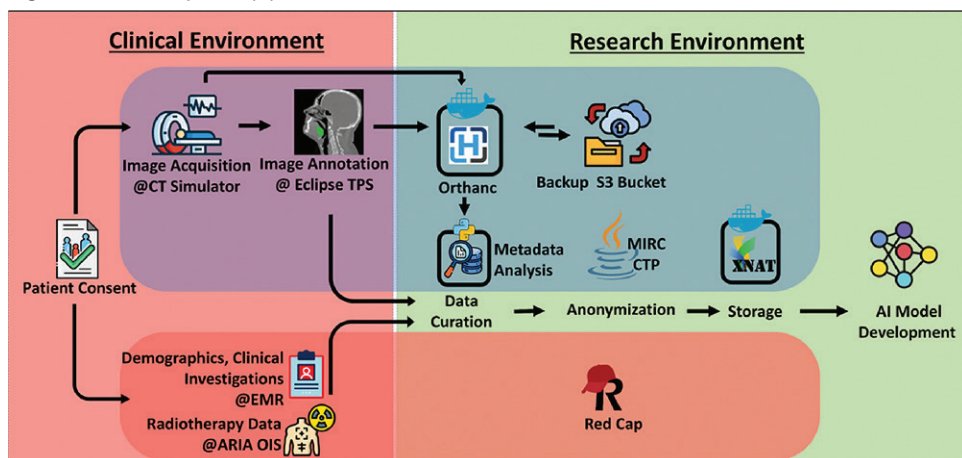
- **Consent and registration:**
Consent processes were revised to include secondary AI-related data use, long-term storage, and controlled sharing, using plain-language explanations. This supports autonomy, transparency, and informed participation.

- Standardised image acquisition:**
 Imaging protocols were harmonised for slice thickness, reconstruction kernels and contrast timing, and radiology where feasible across scanners. Site-specific checklists and technologist training were introduced to reduce avoidable variability. Residual heterogeneity remains and is documented.
- Annotation and peer review:**
 Disease-site specific contouring templates and labelling conventions for tumour and organs at risk were standardised. A subset of cases underwent double reading with consensus to characterise inter-observer variability.
- Automated quality control:**
 Automated scripts screened DICOM data for missing series, inconsistent parameters, and incomplete structure sets, flagging cases for human review prior to dataset inclusion.
- De-identification and storage:**
 Vendor-neutral Orthanc DICOM servers applied configurable de-identification profiles before forwarding images to curated archives with role-based access. Structured clinical datasets were stored in controlled databases on Recap with audit trails.
- Follow-up data:**
 Teleconsultation appointments were scheduled at discharge to support follow-up capture. Follow-up completeness increased over time; however, this change occurred alongside multiple workflow interventions and should not be attributed to a single component.

Scale of deployment

Since implementation of the integrated EMR and imaging pipeline, more than 2,500 radiation oncology electronic records have

Figure 02: AI-ready data pipeline



been captured across two campuses. The infrastructure has supported multiple imaging-based and clinical data projects across disease sites.

Orthanc and XNAT deployments have supported increasing data volumes under current institutional conditions. Extension to multi-institutional networks is technically feasible but would require additional governance agreements, interoperability alignment, and regulatory approvals. Current evidence reflects single-institution, multi-campus scale.

Identified users and beneficiaries

Three principal user groups are involved:

- **Data producers:**
Clinicians, physicists, and therapists generating structured clinical and imaging data within operational systems.

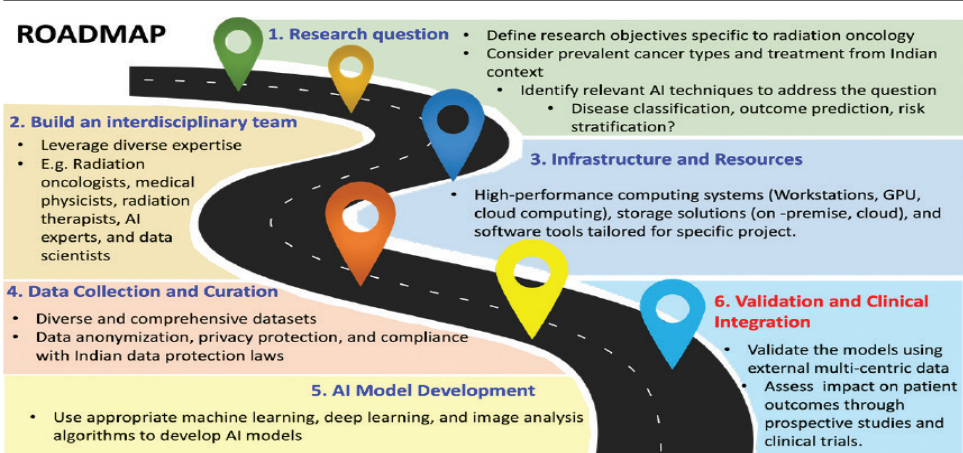
- **Data stewards:**
IT teams, physicists, and data managers responsible for storage, de-identification, security, and access control.
- **Data consumers:**
Approved clinicians, researchers, trainees, and data scientists using curated datasets for analysis, model development, education, and quality improvement.

Patients are indirect beneficiaries through improved data quality and governance supporting future decision-support tools.

Impact

The primary impacts have been infrastructural and organizational rather than algorithmic. Once established, the pipeline (Figure 3) was observed to be reusable and scalable, supporting additional disease sites and AI

Figure 03: Roadmap highlighting components that are required for implementation of AI-projects





use cases with minimal reconfiguration, thereby potentially reducing marginal time and cost for new research efforts (Inampudi et al., 2024). Augmented follow-up data capture through teleconsultation workflows was associated with an increase in outcome data completeness from approximately 30% to 80% over four years. Governance and auditability were strengthened through standardized de-identification profiles, role-based access controls, and comprehensive logging, supporting transparency and accountability principles reflected in global AI ethics guidance. As individual models mature, their reporting can align with frameworks such as TRIPOD+AI, while benefiting from datasets generated through a pipeline already designed to meet reproducibility and data stewardship requirements

Ethics and governance

Ethics and governance were treated as core design requirements. Institutional ethics committees reviewed protocols addressing secondary use, cross-border collaboration, and long-term retention. Governance measures address human oversight, accountability, transparency, data protection, and inclusiveness. Technical safeguards include de-identification, project-level permissions, secure infrastructure, and audit trails. Oversight is provided by a cross-disciplinary steering group. Consent materials clarify that AI tools developed from these data are intended to support, not replace, clinician decision-making.

Discussion

Successes

Prioritising data pipelines before model development yielded a reusable institutional asset supporting multiple projects. Embedding structured capture within routine workflows reduced reliance on parallel research systems. Standards-based and open platforms supported interoperability and cost control. Stakeholder engagement and training supported adoption and data stewardship culture.

Challenges

Implementation required workflow negotiation to integrate structured consent, imaging checks, and documentation standards without disrupting high-volume services. Infrastructure variability and resource constraints complicated harmonisation. Infrastructure work was less visible than model development, making funding and recognition more challenging. Balancing research access with privacy and regulatory limits required ongoing governance alignment.

Lessons learned

Key lessons include treating data stewardship as a primary ethical safeguard, building modular pipelines, preferring interoperable standards, investing in local champions, and aligning digital capture with clinical workflows. AI readiness should be considered a staged infrastructure and governance journey rather than a single technical project.

Conclusion

This experience indicates that AI readiness in radiation oncology depends primarily on structured data pipelines, interoperable systems, governance safeguards, and human capacity rather than algorithm development alone. The framework described here was implemented incrementally within routine workflows and supports production of curated datasets for AI research and evaluation.

The approach is context-specific and requires adaptation to local governance, infrastructure, and regulatory environments. Sustainable AI in clinical settings is supported by deliberate investment in data infrastructure and stewardship, enabling future decision-support tools to be developed and assessed responsibly.

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